## 1. FIRST Set Algorithm

The FIRST set of a symbol is the set of terminals that can begin a string derived from that symbol.

- **Terminal Rule:** If the symbol is a terminal, its FIRST set contains only itself.
- **Non-Terminal Rule:** If the symbol is a non-terminal $A$, we examine all productions $\boldsymbol{A} \rightarrow \boldsymbol{Y_1 Y_2 \ldots Y_n}$.
- **Epsilon Handling:** If $\boldsymbol{A} \rightarrow$ #(epsilon) is a production, add # to $\boldsymbol{FIRST(A)}$.
- **Derivation Chain:** For a production $\boldsymbol{A} \rightarrow \boldsymbol{Y_1 Y_2 \ldots Y_n}$, we compute $\boldsymbol{FIRST(Y_1)}$. If $\boldsymbol{FIRST(Y_1)}$ contains #, we continue to $\boldsymbol{Y_2}$, and so on.

## 2. FOLLOW Set Algorithm

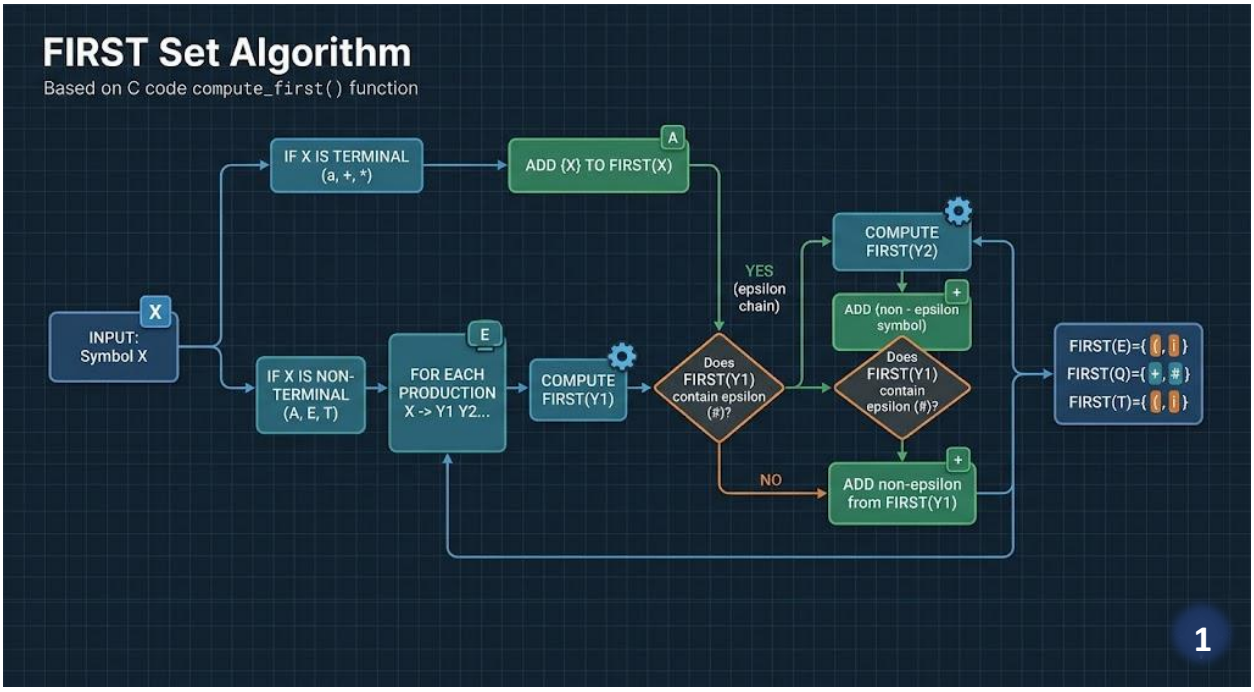
The FOLLOW set of a non-terminal $\boldsymbol{A}$ is the set of terminals that can appear immediately to the right of $\boldsymbol{A}$ in any sentential form.

- **Start Symbol Rule:** The start symbol always includes the end-marker $ in its FOLLOW set.
- **Scanning RHS:** To find $\boldsymbol{FOLLOW(B)}$, we search the right-hand side of all productions for the occurrence of $\boldsymbol{B}$.
- **Immediate Neighbor Rule:** In a production $\boldsymbol{A} \rightarrow \boldsymbol{\alpha B \beta}$, everything in $\boldsymbol{FIRST(\beta)}$ (except #) is added to $\boldsymbol{FOLLOW(B)}$.
- **Recursive Rule:** In a production $\boldsymbol{A} \rightarrow \boldsymbol{\alpha B}$ or $\boldsymbol{A} \rightarrow \boldsymbol{\alpha B \beta}$ where $\boldsymbol{FIRST(\beta)}$ contains #, everything in $\boldsymbol{FOLLOW(A)}$ is added to $\boldsymbol{FOLLOW(B)}$.

## 3. LL(1) Table Construction Algorithm

The table $\boldsymbol{M[A, a]}$ tells the parser which production to use for a non-terminal $\boldsymbol{A}$ and lookahead terminal $\boldsymbol{a}$.

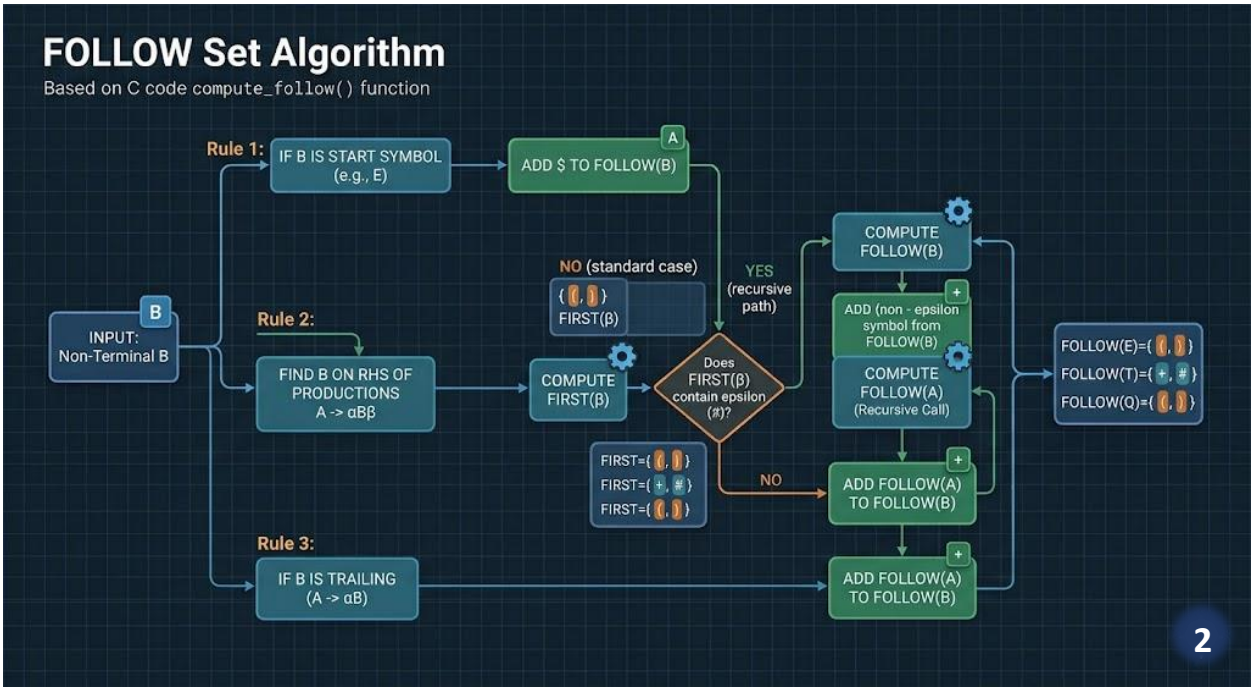
- **Rule 1 (Using FIRST):** For every terminal $\boldsymbol{a}$ in $\boldsymbol{FIRST(RHS)}$, add the production $\boldsymbol{A} \rightarrow \boldsymbol{RHS}$ to $\boldsymbol{M[A, a]}$.
- **Rule 2 (Using FOLLOW):** If # is in $\boldsymbol{FIRST(RHS)}$, then for every terminal $\boldsymbol{b}$ in $\boldsymbol{FOLLOW(A)}$, add $\boldsymbol{A} \rightarrow \boldsymbol{RHS}$ to $\boldsymbol{M[A, b]}$.
- **Rule 3 (EOF):** If # is in $\boldsymbol{FIRST(RHS)}$ and $ is in $\boldsymbol{FOLLOW(A)}$, add $\boldsymbol{A} \rightarrow \boldsymbol{RHS}$ to $\boldsymbol{M[A, \$]}$.




This visual shows the full flow of **`compute_first`**, including the decision points for recursion and handling epsilon (#) derivations in production rules.

**Key Takeaways:**

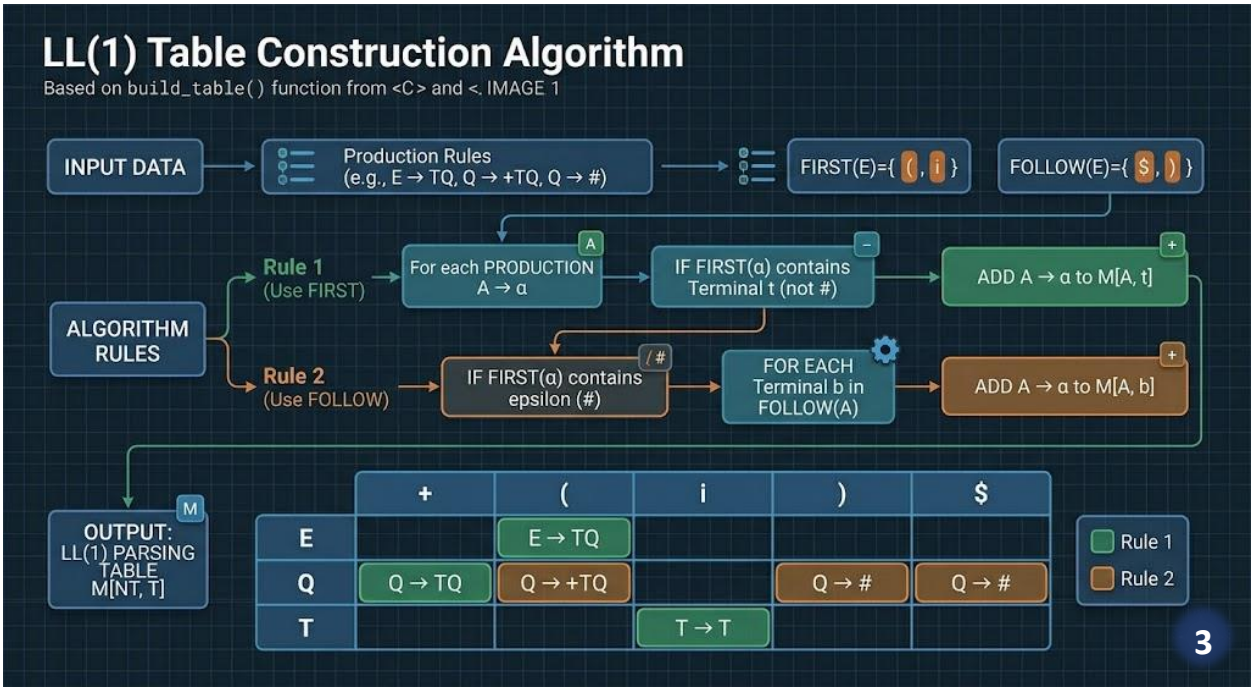
- **Terminals (a, +, …):** Add only the terminal itself to its FIRST set.
- **Epsilon (#):** If a non-terminal derives #, this is added to its FIRST set.
- **Nullable Chain:** If $\boldsymbol{FIRST}(\boldsymbol{Y_1})$ contains #, the algorithm must look at $\boldsymbol{FIRST}(\boldsymbol{Y_2})$, and potentially further down the string, collecting terminals along the way.




This visual detail the **`compute_follow`** flow, explicitly showing the interaction with FIRST sets and recursive calls, which is a major point of collaboration between your two modules.

**Key Takeaways:**

- **Start Symbol Rule:** The start symbol ($\boldsymbol{S}$, or $\boldsymbol{E}$ in this example) always gets $ in its FOLLOW set.
- **Right Neighbors (Rule 2):** For a production $\boldsymbol{A} \rightarrow \boldsymbol{\alpha B \beta}$, add $\boldsymbol{FIRST}(\boldsymbol{\beta})$ (excluding #) to $\boldsymbol{FOLLOW}(\boldsymbol{B})$. This step visually refers to the FIRST sets calculated previously (Image 1).
- **Trailing Non-Terminals (Rule 3):** If $\boldsymbol{A} \rightarrow \boldsymbol{\alpha B}$ or if $\boldsymbol{FIRST}(\boldsymbol{\beta})$ contains #, add $\boldsymbol{FOLLOW}(\boldsymbol{A})$ to $\boldsymbol{FOLLOW}(\boldsymbol{B})$, often requiring recursive calls.




This visual shows the multi-stage process of taking the FIRST sets (Image 1), the FOLLOW sets (Image 2), and applying the rules to populate the final parsing table structure, perfectly organized.

**Key Takeaways:**

- **Rule 1 (Use FIRST):** For each terminal $\boldsymbol{a}$ in $\boldsymbol{FIRST(RHS)}$, add the production $\boldsymbol{A \rightarrow RHS}$ to entry $\boldsymbol{M[A, a]}$. This step relies heavily on FIRST sets (Image 1).
- **Rule 2 (Use FOLLOW):** If # (epsilon) is in $\boldsymbol{FIRST(RHS)}$, then for each terminal $\boldsymbol{b}$ in $\boldsymbol{FOLLOW(A)}$, add $\boldsymbol{A \rightarrow RHS}$ to $\boldsymbol{M[A, b]}$. This step relies on FOLLOW sets (Image 2).
- **EOF Rule:** If # is in $\boldsymbol{FIRST(RHS)}$ and $ is in $\boldsymbol{FOLLOW(A)}$, add $\boldsymbol{A \rightarrow RHS}$ to $\boldsymbol{M[A, \$]}$.
- **Table Structure:** The resulting table, perfectly organized with different color highlights for Rule 1 and Rule 2, visually maps each production to its unique parsing cell.